

1 Sketches of the System

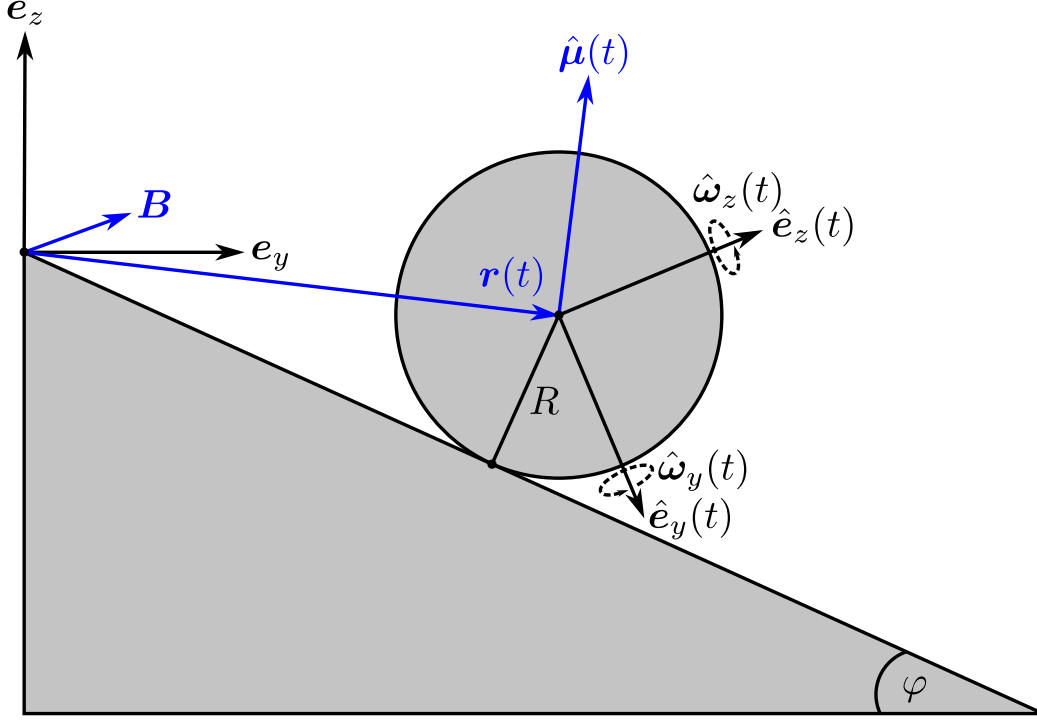


Figure 1: Depicted is the yz -projection of the incline system, the z -direction is orthogonal to the paper plane. The incline has an inclination angle φ and the sphere is of radius R . Two orthonormal coordinate frames are considered: The fixed frame $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$ and the body frame $\{\hat{\mathbf{e}}_x, \hat{\mathbf{e}}_y, \hat{\mathbf{e}}_z\}$. Vectors expressed in the body frame carry hats and are time-dependent, stationary vectors not. Blue vectors might have a non-zero z -component and are therefore only projections. The outer magnetic field is represented by $\mathbf{B}$, the magnetic dipole vector of the sphere by $\hat{\boldsymbol{\mu}}(t)$, and the centre of mass measured from the fixed frame by $\mathbf{r}(t)$. The in-plane angular velocities of the body frame are shown as $\hat{\omega}_x$ and $\hat{\omega}_y$.

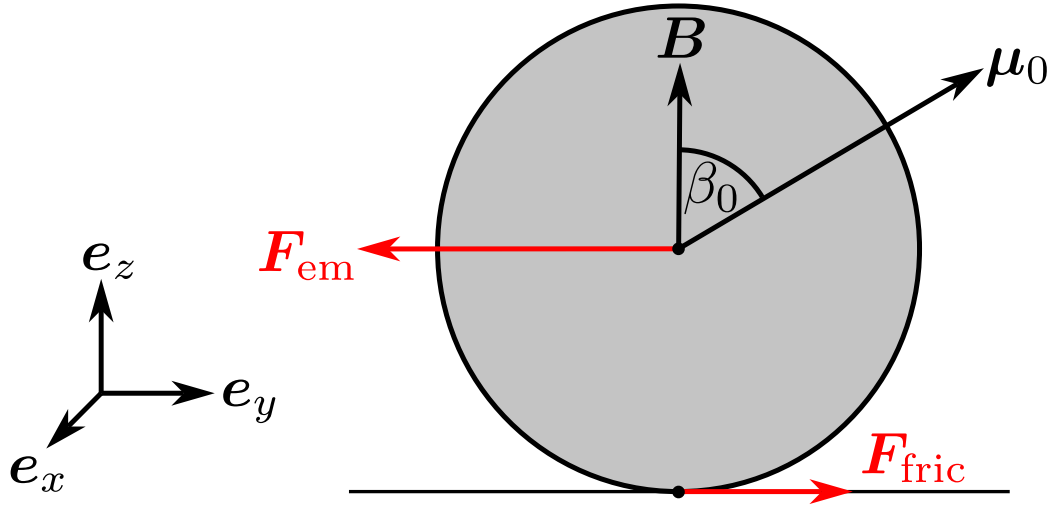


Figure 2: Shown is a sketch of the flat system with sufficiently small initial displacement angle β_0 between the outer magnetic field B and the initial magnetic dipole vector μ_0 . Note that the system is effectively reduced to two dimensions. The red vectors indicate the oscillation-driving electromagnetic force F_{em} and the damping frictional force F_{fric} . The frictional force includes aerodynamical drag and rolling resistance.

2 Figure Family A

These figures are results of a proof-of-concept toy model: The incline without electromagnetic interaction. The goal is to compare numerical results to analytical equations.

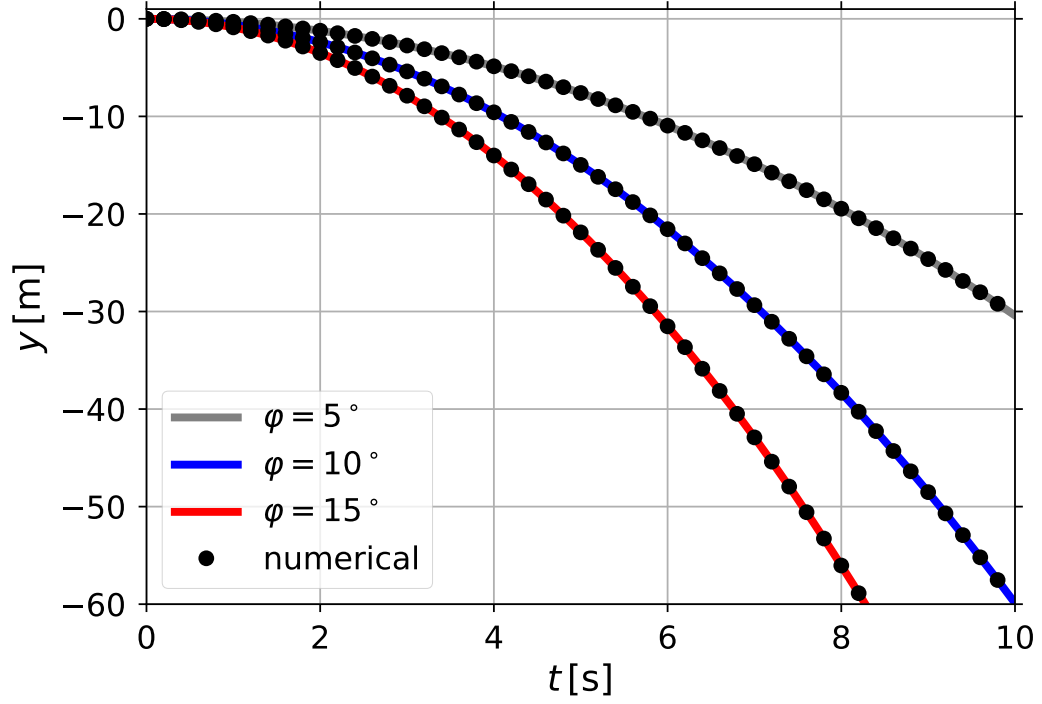


Figure A1: Shown are different evolutions of the y -projection of the centre-of-mass coordinate $\mathbf{r} = (x, y, z)^\top$ over time t for different incline angles φ without friction. Lines were calculated from the analytical solution and compared to the numerical results, represented as dots.

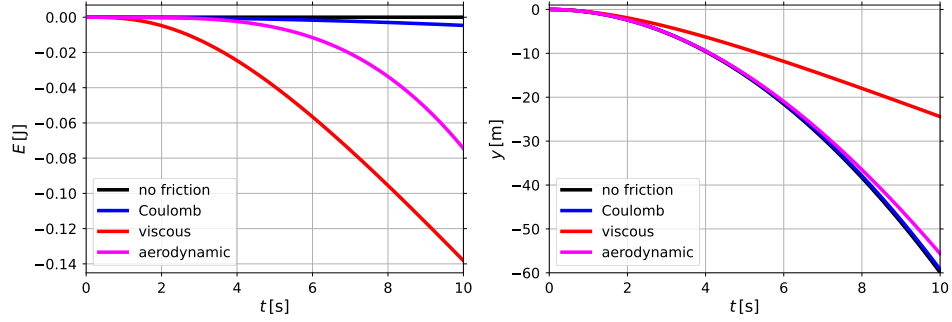


Figure A2/A3: The left plot shows the energy evolution $E(t)$ of the system over time t , while the right one depicts the y -COM-coordinate evolution. The data originates from the numerical simulation program. Compared are different friction models, where *Coulomb* represents dry rolling resistance, *viscous* represents a linear damping approximation to rolling resistance, and *aerodynamic* represents spherical aerodynamic drag.

3 Figure Family B

4 Figure Family C

5 Figure Family D

The following figures are the results of running the simulation for $\Delta t = 1\text{s}$ simulation time with an incline of $\varphi = 10^\circ$ for all possible values of $\mu_x(t=0)$ and $\mu_y(t=0)$ in $[-1, 1]$ for increments of 0.1, meaning these are the results of 441 calculations.

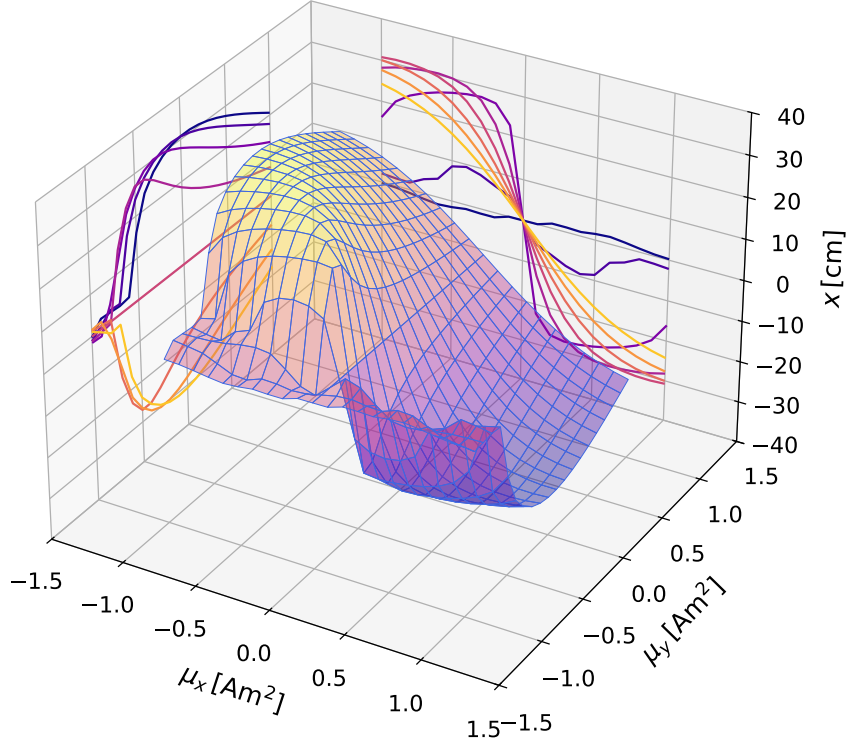


Figure D1: The surface represents the x -value the sphere attains after $t = 1$ s as a function of the initial x - and y -components of the magnetic dipole vector of the sphere. Since the x -direction is transverse to the rolling motion induced by gravity, it can be seen as a measure for the influence of the magnetic force on the trajectory. The graphs at the $\mu_y x$ -plane are slices of the surface for fixed values of μ_x , and respectively for the graphs on the $\mu_x x$ -plane with fixed μ_y .

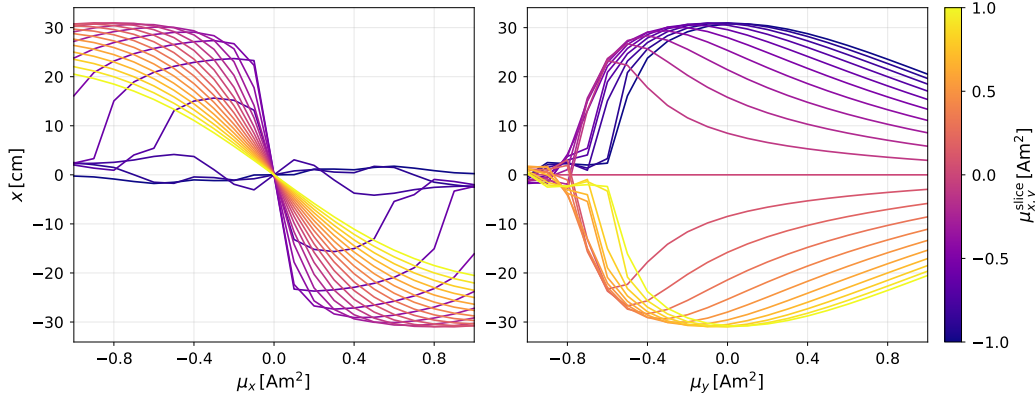


Figure D2: These plottings are enhanced versions of the slices of figure D1, left for different fixed μ_y and right for different fixed μ_x . Keeping μ_y fixed produces symmetrical deviations from the free rolling path depending on sign and magnitude of μ_x , as one would expect. A clear transition from regular sigmoid behaviour to non-smooth responses occurs at highly negative values of μ_y , at which the ball no longer rolls down, discussed more rigorously in D3. The slice graphs for fixed μ_x on the right show a characteristic constant curve for $\mu_x = 0$, as there can be no deviation from the purely gravitational path without a torque non-parallel to the yz -plane. All other slices display an asymmetry for positive and negative values. This is explained by the fact that for positive μ_y , the direction of rolling increases the misalignment of $\boldsymbol{\mu}$ with $\boldsymbol{B}$, slowing down the motion and hence also the deviation. On the other hand, negative μ_y -values cause $\boldsymbol{\mu}$ to align with $\boldsymbol{B}$, accelerating the motion in general and hence also the deviation in particular. A less regular asymmetry occurs for larger negative μ_y -values, which is again the critical region where the magnetic force dominates over and acts against gravity, causing looping paths instead of downward motion.

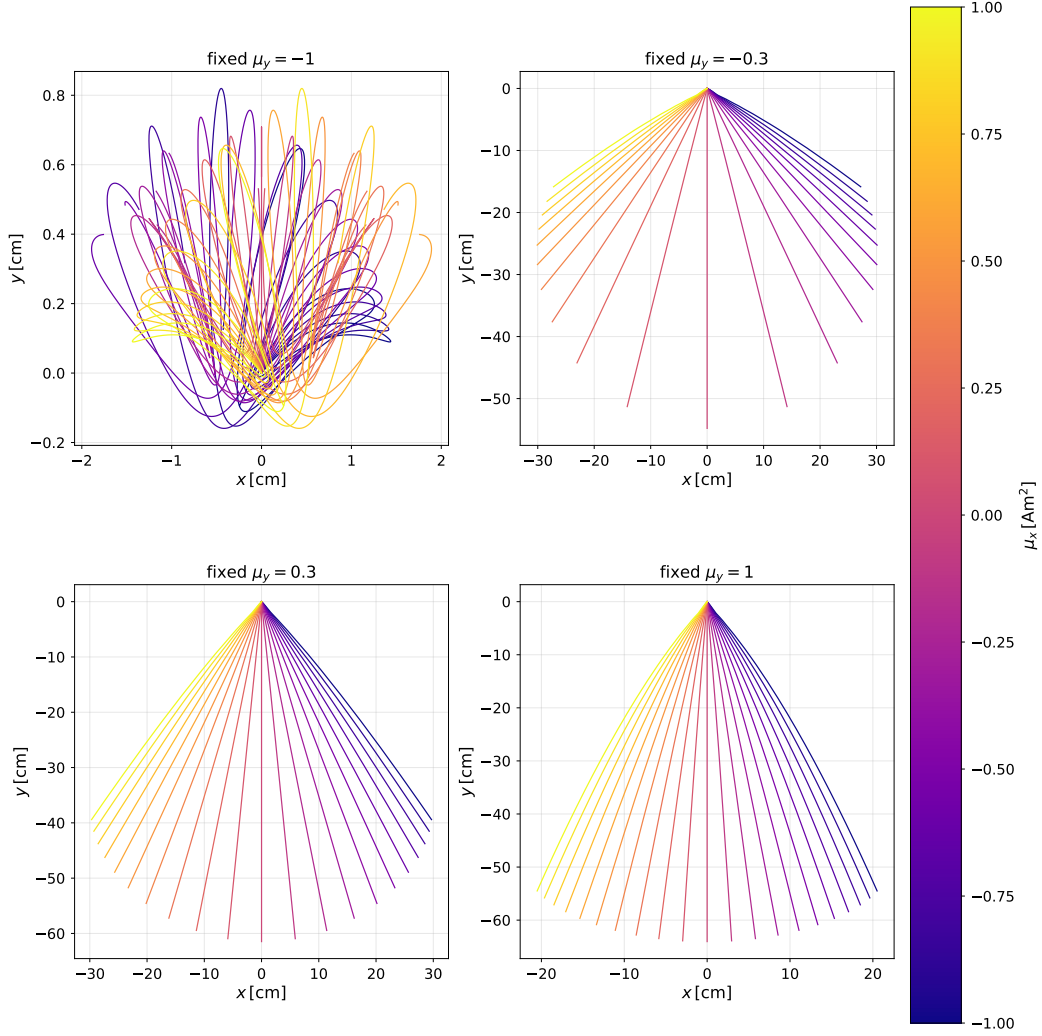


Figure D3: Visible are different trajectories of the sphere on the xy -plane for selected μ_y -slices and all μ_x -values corresponding to these slices. Here, the anomaly in D1 and D2 for large negative μ_y -values can be seen immediately. For smaller negative and all positive μ_y -slices, μ_y only controls the spread of the trajectories, while μ_x influences the deviation from a straight path. At $\mu_y = -1$, the magnetic force induced by the μ_y -component as soon as the μ_x -induced torque bends the trajectory out of the yz -plane overcompensates gravity and causes an upward motion. After that, the sphere moves in roughly elliptic loops.

6 Figure Family E

The following figure shows the behaviour of the y -coordinate for different values of $\mu_z(t=0)$ in $[0, 1]$, incremented by 0.01-steps. The goal was to see the transition point at which the magnetic force overcomes the gravitational force, so $\mu_x(t=0) = \mu_y(t=0) = 0$.

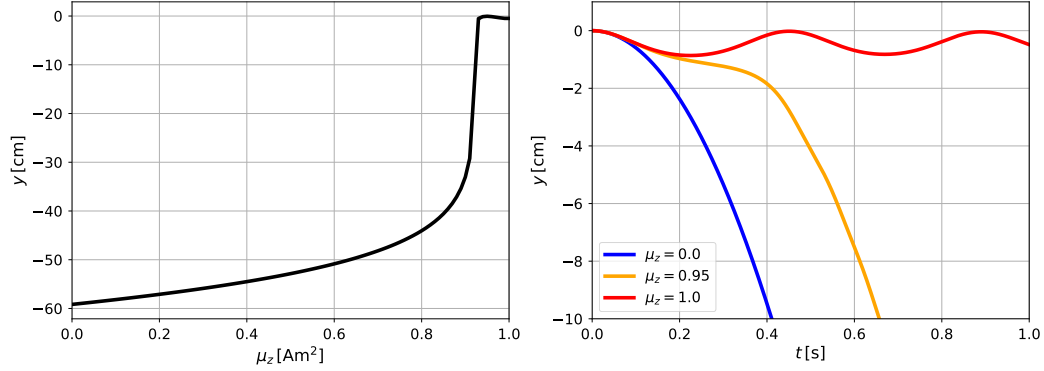


Figure E: The graph on the left shows the y -projection of the sphere's COM at $t = 1$ s as a function of initial μ_z -values. There are roughly three discernible regions: First, the y -value decreases, which is expected as the rolling motion increases the misalignment of $\boldsymbol{\mu}$ with $\boldsymbol{B}$. On the right, the y -trajectory for $\mu_z = 0$ is representative for this region. For increasingly high μ_z -values, a transition phase occurs. The representing graph at $\mu_z = 0.95$ shows that in this region, the sphere begins to oscillate slightly at first, but gravity soon overcomes the magnetic force. For $\mu_z \geq 0.96$, the magnetic force fully overcomes the gravitational force which only causes small oscillations around the origin, as seen by the representing graph for $\mu_z = 1$.